

The arithmetic mean disproves the universal bounded-affine lower bound in Conjecture 11

Candidate counterexample packet for arXiv:2110.04763

23 July 2026

Status. Candidate counterexample, likely valid, pending human review. The result concerns Conjecture 11 exactly as printed. It does not address a possible intended restriction to the maximum aggregation.

1 Source and target

Attias and Kontorovich study fat-shattering dimension under pointwise aggregation of real-valued function classes [1]. Their Theorem 3 gives a dimension-free upper bound for Lipschitz aggregations of bounded affine classes. In Section 5, page 15 of the arXiv PDF, they state Conjecture 11: for R_i -bounded affine classes F_i and an aggregation rule G that is 1-Lipschitz in supremum norm,

$$\text{fat}_\gamma(G(F_1, \dots, F_k)) \geq \frac{C \text{Log}(k)}{\gamma^2} \sum_{i=1}^k R_i^2, \quad 0 < \gamma < \min_i R_i, \quad (1)$$

where $C > 0$ is universal. The same wording appears in the final JMLR version as Conjecture 11, equation (29).

2 Definitions

Let H be a real Hilbert space and let $B_H = \{x \in H : \|x\| \leq 1\}$. For $R > 0$, write

$$\mathcal{A}_R(H) = \{x \mapsto \langle w, x \rangle + b : w \in H, \|w\| \leq R, |b| \leq R\}.$$

This is precisely the equal-radius instance of the bounded affine classes used in the source paper (and includes its finite-dimensional setting).

A finite set $\{x_1, \dots, x_m\}$ is γ -shattered by a function class F if there are thresholds $r_1, \dots, r_m$ such that for every sign vector $\varepsilon \in \{-1, 1\}^m$ there is $f_\varepsilon \in F$ with

$$\varepsilon_i(f_\varepsilon(x_i) - r_i) \geq \gamma \quad (i = 1, \dots, m).$$

The largest possible m is $\text{fat}_\gamma(F)$.

For $k \geq 1$, let

$$M_k(z_1, \dots, z_k) = \frac{1}{k} \sum_{j=1}^k z_j$$

be the arithmetic-mean aggregation.

5. Discussion

In this paper, we proved upper and lower bounds on the fat-shattering dimension of aggregation rules as a function of the fat-shattering dimension of the component classes. We leave some remaining gaps for future work. First, for aggregation rules that commute with shifts, assuming $\text{fat}_\gamma(F_i) \leq d$, for $1 \leq i \leq k$, we show in Theorem 1 that

$$\text{fat}_\gamma(G(F_1, \dots, F_k)) \leq Cdk \log^2(dk), \quad \gamma > 0,$$

$C > 0$ is a universal constant. We pose the following

Open problem. Let G be an aggregation rule that commutes with shifts. Is it the case that for all $F_i \subseteq \mathbb{R}^\Omega$ with $\text{fat}_\gamma(F_i) \leq d$, $i \in [k]$, we have

$$\text{fat}_\gamma(G(F_1, \dots, F_k)) \leq Cdk \log(k), \quad \gamma > 0,$$

for some universal $C > 0$?

In light of Theorem 8, this is the best one could hope for in general. We pose also the following conjecture about bounded affine functions.

Conjecture 11 *Theorem 3 is tight up to constants. For R_i -bounded affine functions and aggregation rule G that is 1-Lipschitz in supremum norm,*

$$\text{fat}_\gamma(G(F_1, \dots, F_k)) \geq \frac{C \text{Log}(k)}{\gamma^2} \sum_{i=1}^k R_i^2, \quad 0 < \gamma < \min_{i \in [k]} R_i, \quad (22)$$

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Figure 1: Section 5, page 15 of the source PDF. The target is Conjecture 11 and its displayed universal lower bound (22).

3 Proof intuition

The proposed lower bound charges a term of order $R_i^2 \text{Log}(k)$ for every input class, regardless of how the aggregation uses those inputs. Averaging identical convex affine classes does not create any new functions: averaging their parameters stays in the same parameter ball, and every original function is recovered by putting it in all k coordinates. Thus the aggregate class is independent of k .

For completeness, its fat dimension is bounded without invoking any external entropy theorem. Average the shattering inequalities over random signs. The arbitrary threshold vector cancels in expectation, while the Hilbert norm and intercept contributions are each at most $R\sqrt{m}$. This forces $m \leq 4R^2/\gamma^2$, so the conjectured $k \text{Log}(k)$ growth is impossible.

4 Counterexample

Lemma 1 (Fat dimension of a bounded affine class). *For every real Hilbert space H , $R > 0$, and $\gamma > 0$,*

$$\text{fat}_\gamma(\mathcal{A}_R(H)) \leq \frac{4R^2}{\gamma^2}.$$

Proof. Suppose $x_1, \dots, x_m \in B_H$ are γ -shattered with thresholds $r_1, \dots, r_m$. For each sign vector $\varepsilon \in \{-1, 1\}^m$, choose $f_\varepsilon(x) = \langle w_\varepsilon, x \rangle + b_\varepsilon \in \mathcal{A}_R(H)$ witnessing its labeling. Summing the m shattering

inequalities gives

$$m\gamma \leq \sum_{i=1}^m \varepsilon_i (f_\varepsilon(x_i) - r_i) \leq \sup_{\substack{\|w\| \leq R \\ |b| \leq R}} \sum_{i=1}^m \varepsilon_i (\langle w, x_i \rangle + b - r_i).$$

The supremum on the right is exactly

$$R \left\| \sum_{i=1}^m \varepsilon_i x_i \right\| + R \left| \sum_{i=1}^m \varepsilon_i \right| - \sum_{i=1}^m \varepsilon_i r_i.$$

Now take expectation over independent uniform signs. The last term has expectation zero. Therefore

$$\begin{aligned} m\gamma &\leq R \mathbb{E} \left\| \sum_{i=1}^m \varepsilon_i x_i \right\| + R \mathbb{E} \left| \sum_{i=1}^m \varepsilon_i \right| \\ &\leq R \left(\mathbb{E} \left\| \sum_{i=1}^m \varepsilon_i x_i \right\|^2 \right)^{1/2} + R \left(\mathbb{E} \left| \sum_{i=1}^m \varepsilon_i \right|^2 \right)^{1/2} \\ &= R \left(\sum_{i=1}^m \|x_i\|^2 \right)^{1/2} + R\sqrt{m} \leq 2R\sqrt{m}. \end{aligned}$$

For $m > 0$, division by $\sqrt{m}\gamma$ and squaring yield the claim. $\square$

Theorem 1 (Arithmetic-mean counterexample). *For every $k \geq 1$,*

$$M_k(\mathcal{A}_R(H), \dots, \mathcal{A}_R(H)) = \mathcal{A}_R(H),$$

and $M_k : \mathbb{R}^k \rightarrow \mathbb{R}$ has optimal Lipschitz constant 1 with respect to the supremum norm. Consequently, no universal $C > 0$ can make (1) hold for all k , all bounded affine classes, and all 1-Lipschitz aggregation rules.

Proof. For $z, z' \in \mathbb{R}^k$,

$$|M_k(z) - M_k(z')| \leq \frac{1}{k} \sum_{j=1}^k |z_j - z'_j| \leq \|z - z'\|_\infty.$$

Equality occurs when every coordinate difference is the same nonzero number, so the optimal Lipschitz constant is exactly 1.

If $f_j(x) = \langle w_j, x \rangle + b_j \in \mathcal{A}_R(H)$, then

$$M_k(f_1, \dots, f_k)(x) = \left\langle \frac{1}{k} \sum_{j=1}^k w_j, x \right\rangle + \frac{1}{k} \sum_{j=1}^k b_j.$$

The averaged parameters still satisfy

$$\left\| \frac{1}{k} \sum_j w_j \right\| \leq R, \quad \left| \frac{1}{k} \sum_j b_j \right| \leq R.$$

This proves $M_k(\mathcal{A}_R(H), \dots, \mathcal{A}_R(H)) \subseteq \mathcal{A}_R(H)$. Conversely, every $f \in \mathcal{A}_R(H)$ is the mean of the diagonal tuple $(f, \dots, f)$, proving equality.

Now set $F_i = \mathcal{A}_R(H)$ and $R_i = R$ for every i . Lemma 1 gives

$$\text{fat}_\gamma(M_k(F_1, \dots, F_k)) = \text{fat}_\gamma(\mathcal{A}_R(H)) \leq \frac{4R^2}{\gamma^2}.$$

But (1) would give

$$\text{fat}_\gamma(M_k(F_1, \dots, F_k)) \geq \frac{Ck \text{Log}(k)R^2}{\gamma^2}.$$

Taking, for example, $\gamma = R/2$, the first bound is at most 16, whereas the second is $4Ck \text{Log}(k)$, which exceeds 16 for all sufficiently large k . This contradicts every proposed universal $C > 0$. $\square$

5 Interpretation and limitations

The counterexample uses one of the paper’s own canonical aggregation rules: the mean. It is symmetric, depends on every coordinate, is shift-commuting, and has exact Lipschitz constant one. Thus the obstruction is not an artifact of a constant map or of an aggregation that ignores most inputs.

The conclusion is deliberately narrow. The phrase “Theorem 3 is tight up to constants” may have been intended existentially (some aggregation attains the upper bound) or specifically for $G_{\max}$. The displayed Conjecture 11, however, states the lower bound for an aggregation rule G under only the 1-Lipschitz hypothesis. The arithmetic mean disproves that universal formulation. It says nothing against the max-specific lower bound, which remains outside this packet.

6 Verification and novelty check

The proof has no computational or conditional dependency. The two substantive checks are the exact class identity under averaging and the random-sign estimate in Lemma 1; both are written out above.

Before promotion, the run’s registry and lightweight indexes were searched for arXiv:2110.04763, the exact title, and the core bounded-affine aggregation terms. No duplicate result appeared. The current arXiv source and the official 2024 JMLR version were inspected; both retain Conjecture 11 in the form shown in Figure 1. Bounded exact-phrase web searches for the title, “Conjecture 11,” “1-Lipschitz” aggregation, and the displayed lower bound found no correction, erratum, or later paper recording this counterexample. This is not an exhaustive novelty guarantee. Because the argument is elementary and may identify an unintended quantifier, novelty confidence is moderate; confidence in the literal mathematical counterexample is high.

7 Human-review recommendation

The recommended review is to confirm the intended quantifier in Conjecture 11. If the printed universal reading is intended, Theorem 1 is a complete counterexample. If the intended claim is max-specific or merely existential sharpness of Theorem 3, the source statement should be narrowed and this packet should be classified as a statement-level counterexample rather than a refutation of that intended variant.

References

- [1] I. Attias and A. Kontorovich, *Fat-Shattering Dimension of k -fold Aggregations*, Journal of Machine Learning Research **25** (2024), paper 144, 1–29; arXiv:2110.04763.